

The Dollarisation Paradox in Cambodia: Network Externalities Matter*

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Abstract

The increase in dollarisation in Cambodia has run counter to the general belief that macroeconomic and political stability help reduce it. We provide the first explanation for this counterfactual phenomenon. In doing so, this paper develops a cash-in-advance model by including a dollar pricing index to amplify the network effects of using a foreign currency (denoted dollar). The dollar pricing index, defined as the share of active prices quoted in dollars and derived from firms' quotation choices, reduces the dollar's transaction costs, thereby increasing its use in the economy. This increased use of the dollar further improves the experience of using it, thereby leading to higher dollar usage in price quotations. The positive interaction between using the dollar as a unit of account and as a means of payment continues to drive dollarisation higher, even though the economy has achieved low inflation and political stability. A disciplined minimum-distance benchmark against a nested Uribe specification further shows that the quotation margin is needed to rationalise Cambodia's divergence from the declining dollarisation paths observed in Laos P.D.R., Vietnam, and Peru.

JEL classification: E41, F41

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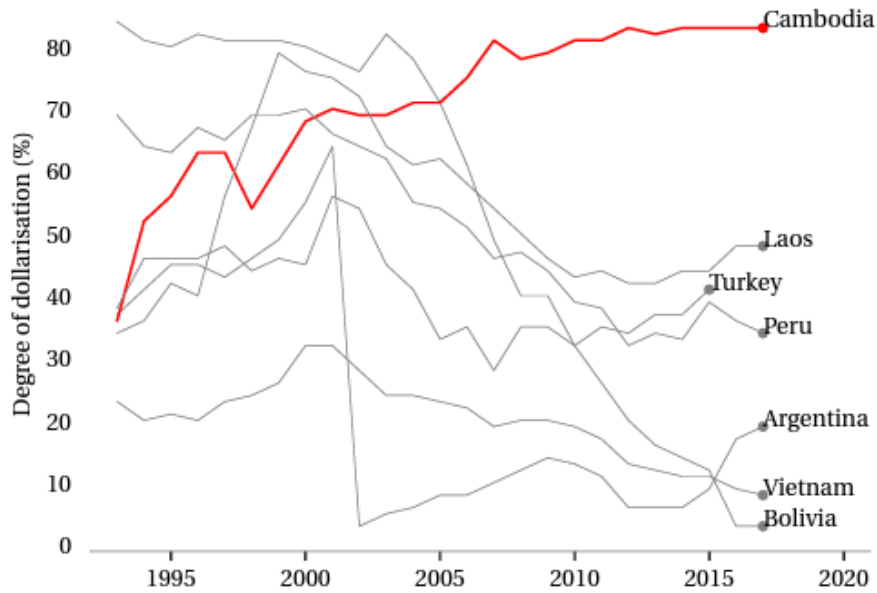
“In Cambodia, the Ghosts Prefer Dollars.”

— The New York Times (2016)

1 INTRODUCTION

The headline in the New York Times (2016) implies that the US dollar is a dominant currency in Cambodia, rather than the Cambodian riel. When a country such as Cambodia adopts a foreign currency as a legal tender (as a result of high inflation), this financial adaptation is commonly known as *dollarisation*. A common empirical pattern is that, after macroeconomic and political stabilisation, many countries experience a decline or a slow unwinding of dollarisation, although Uribe (1997) emphasises that such adjustment can be highly persistent. However, Cambodia appears to be an odd one out among developing countries that have used a foreign currency. Against the backdrop of its macroeconomic and political stability, Cambodia has been increasingly using a foreign currency, whereas its neighbours (Vietnam and Laos P.D.R.) and others have experienced the opposite trends (see Figure 1).

Figure 1: Trends of dollarisation in selected highly dollarised economies



Note: Degree of dollarisation is a ratio of dollar-denominated deposits to broad money (M2). Data are sourced from the International Monetary Fund Country Reports (Appendix A).

Figure 1 is intended as a descriptive background rather than as the full set of country cases analysed later in the paper. The later comparative discussion focuses on Laos P.D.R., Vietnam, and Peru because these countries provide clear policy contrasts in the regulation of foreign-

currency quotation, whereas Turkey, Argentina, and Bolivia are included only to place Cambodia in a broader set of highly dollarised economies.

To shed light on this counterfactual phenomenon in Cambodia, our work provides both a theoretical framework and some evidence. We develop a theoretical model on the basis of Uribe (1997) cash-in-advance framework. Differing from Uribe (1997), we include a new variable which we call *dollar pricing index*¹ and define as a proportion of an economy that quotes the dollar as a unit of account. We model this quotation margin explicitly through firms' currency-of-quotation decisions, so that the dollar pricing index captures government policies dealing with dollarisation and amplifies the network effects of using the dollar in an economy.

The network effects are the central feature in Uribe (1997) model to explain dollarisation dynamics in developing countries.² In his model, Uribe (1997) assumes that an economy's accumulated experience in using the dollar as a medium of exchange acts as an externality that lowers the transaction cost of buying goods with dollars. However, the network effects would be far more significant if dollar quotations were included³ because dollar-denominated goods promote and ease dollar usage. Therefore, the inclusion of the dollar pricing index further reduces the transaction costs of buying goods with dollars, increasing their use in the economy. The greater use of the dollar then improves the knowledge of using it, encouraging the economy to quote prices in dollars. The interaction between the use of the dollar in the price quotation and the accumulated experience of using it increases the degree of dollarisation, even though the economy has achieved social, political, and economic stability.

Two waves of survey data⁴ in Cambodia along with qualitative data from Laos P.D.R., Vietnam, and Peru, support this augmented model's prediction. With the increasing use of foreign currencies as a unit of account in Cambodia, their use as a medium of exchange is also growing. Over the period 2010-2014, the dollar pricing index and the degree of dollarisation have increased from 28% to 43% and 81% to 83%, respectively. This positive relationship was also observed in Cambodia regions. Figure 2 shows that the higher a region's dollar pricing index, the larger the proportion of revenue it receives in foreign currencies.

While Cambodia has no restrictions on which currency may be used to quote prices, Laos

¹To the best of our knowledge, this index, generated from survey data, appears to be available for only the Cambodian economy.

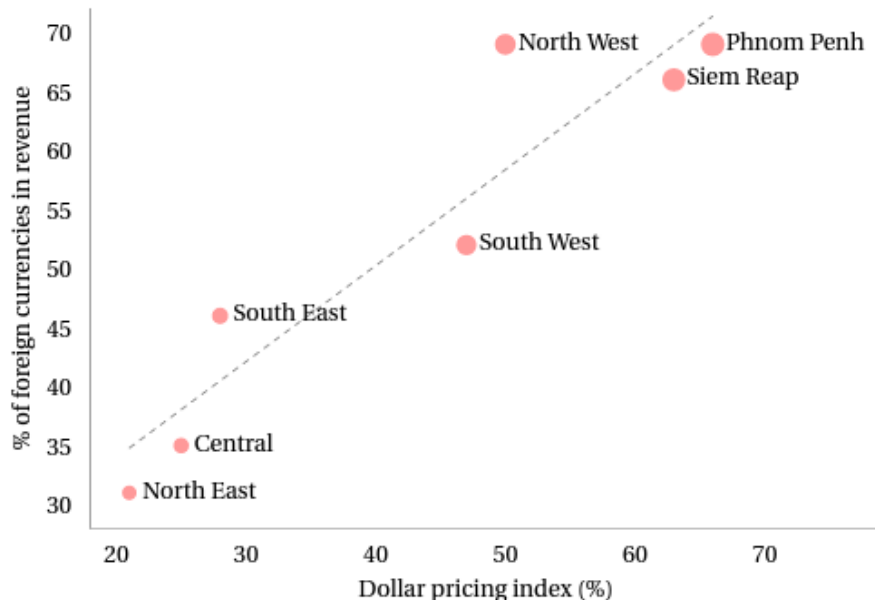
²Uribe (1997) shows that a temporary high inflation episode can generate persistent dollarisation because accumulated experience in using the dollar lowers its transaction cost. Our paper extends this hysteresis framework by allowing firms to explicitly choose the unit-of-account margin and for it to be affected by quotation policy.

³Some dollarised economies, such as Laos P.D.R., Vietnam, and Peru, officially ban a foreign currency from being used as a unit of account, while another like Cambodia does not forbid such a role of foreign currency in its economy.

⁴The first survey was conducted by Khou (2012) in 2010 with a sample of 1106 local merchants, and the second one was conducted by the National Bank of Cambodia (NBC) and JICA research institute in 2014-2015 with a sample of 856 enterprises.

P.D.R., Vietnam, and Peru have banned the use of foreign currencies as a unit of account. The ban has decreased the dollar pricing index, thereby raising the cost of using dollars in transactions, which further reduced their use. As a consequence, the degree of dollarisation in these countries has gradually declined. Section 6 complements this qualitative comparison with a disciplined minimum-distance benchmark in which the four countries share a common structural parameterisation, the policy wedge follows the dated regulatory changes summarised in Table 2, and countries differ in their observed depreciation paths and inherited quotation stocks.

Figure 2: Dollar pricing index and % of foreign currencies in revenue in Cambodia regions



Note: The dollar pricing index is measured by a proportion of products for which merchants quote their prices in foreign currencies. The data are retrieved from the National Bank of Cambodia (2016, p.53-58).

The rest of the paper proceeds as follows. Section 2 reviews the relevant literature concerning currency risk and network externalities as the roots of dollarisation. Section 3 briefly discusses the dollarisation in Cambodia. Section 4 develops a model of currency substitution and compares the extended framework with the Uribe benchmark. Section 5 discusses the effects of government policies, including how changes in depreciation rate (or inflation), dollar pricing index, or both affect the equilibrium. Section 6 provides qualitative evidence and a disciplined minimum-distance benchmark to support the model's predictions, and Section 7 concludes the paper with policy implications and model limitations.

2 CURRENCY RISK AND NETWORK EXTERNALITIES

The present study relates to two main strands of the literature. One concern is how currency risk triggers the use of a foreign currency. The other focuses on the network externalities of using a foreign currency to help explain the evolution of dollarisation in developing countries.

Craig and Waller (2004) and Camera et al. (2004) develop a model in which households use both domestic and foreign currencies in their transactions. Because the domestic currency is perceived to have aas likely to lose value before it is used in a transaction, households tend to use and accept the foreign currency. However, because using the foreign currency incurs a transaction cost,⁵ the domestic one still remains a medium of exchange. This coexistence of both currencies neither supports nor contradicts Gresham's Law, which holds that risky money drives safe money out of the economy (Rolnick and Weber, 1986).

Although the currency risk and transaction cost have opposing effects on the extent of currency substitution, their magnitudes of impact differ. The studies by Melvin and Fenske (1992) and Clements and Schwartz (1993) show that the effects of a decrease in currency risk are weaker in economies that have already experienced exchange rate instability, suggesting that the effects may be slow to emerge in such economies.

Based on the idea that using a foreign currency incurs a transaction cost, Uribe (1997) develops a simple cash-in-advance model, to which this paper is the closest. In his model, the network externalities play a central role in explaining the continued existence of the foreign currency in an economy once it has already been widely used as a medium of exchange. In other words, the more experience an economy has with using the foreign currency as a medium of exchange, the lower its transaction cost becomes. The present paper builds on this hysteresis logic, but adds a separate firm-side decision over the currency in which prices are quoted and allows government policy to act directly on that quotation margin. These network effects are also present in the search-theoretic models of money (see, e.g., Matsuyama et al., 1993; Trejos and Wright, 1995). Kiyotaki and Wright (1989) show that even an item without any intrinsic value can become a means of payment if widely accepted as such. Then, the use of such a thing fundamentally becomes the norm. This indicates that if a foreign currency is broadly used during a period of high inflation, its use will continue after inflation returns to a lower level.

Empirical evidence for network effects on hysteresis has been reported in several studies. For instance, Valev (2010) uses survey data to examine the impacts of currency risk and network externalities on the hysteresis of currency substitution in Bulgaria. He finds the network externalities to be a major factor of hysteresis (see also Duffy and Ochs, 2002).

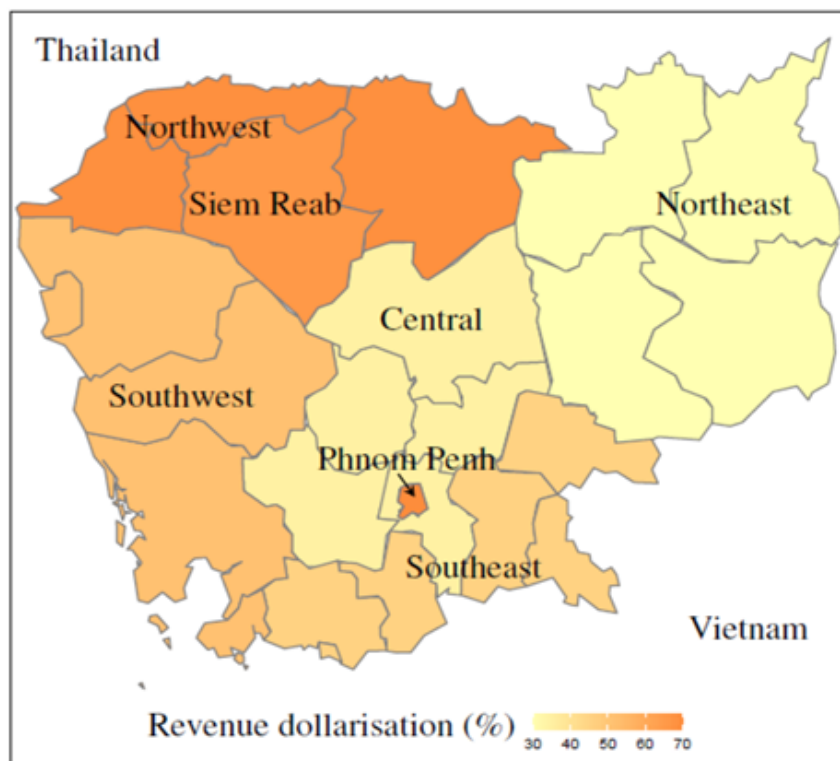
⁵The inclusion of the transaction cost in the model is motivated by the fact that in developing countries, households might need to verify the authenticity of an unfamiliar currency (Engineer, 2000; Tandon, 2003), and spend time switching between currencies (Guidotti, 1992).

3 THE DOLLARISATION IN CAMBODIA

The currency risk and network externalities are the roots of rising dollarisation in Cambodia. Cambodians had already lost trust in the domestic currency (riel) when the riel and economic institutions (e.g. banks), among others, were abolished during the Khmer Rouge regime, known as the Cambodia Genocide, from 1975 to 1979. In March 1980, the riel was reintroduced into the economy. Yet, people preferred barter and gold for most domestic transactions (De Zamaroczy and Sa, 2002). This loss of confidence was further eroded by a large depreciation of the riel during the period 1988-1991.

In the early 1990s, the use of the US dollar grew rapidly as Cambodia opened to foreign investment and aid. The United Nations Transitional Authority spent around USD 1.7 billion – roughly 75% of Cambodia’s GDP at that time – on rents and local services for its peacekeeping operations, including assisting Cambodia with its first national election in 1993 (National Bank of Cambodia, 2016, p.8).

Figure 3: The use of foreign currencies across regions in Cambodia



Note: Revenue dollarisation is a fraction of foreign currencies in revenues. The US dollar is used predominantly in two cities – Phnom Penh and Siem Reap. The US dollar and Thai currency (baht) are widely used along the northeast and southwest border with Thailand. The US dollar and the Vietnamese dong are frequently used along the US-Vietnam border. In other regions, which are mainly rural, the domestic currency (riel) is predominant. Source: National Bank of Cambodia (2016, p.53–58).

For the sake of price stability and economic growth, Cambodia has placed no restrictions on the use of foreign currencies, leading to the use of multiple currencies. The type of currency used as a means of payment is positively associated with the type of currency used as a unit of account. For instance, as merchants in Phnom Penh City use the US dollar to display prices for approx 65% of their total products, they also receive around 69% of their revenues in US dollars (Figure 2). Moreover, the type of currency used in price quotation depends mainly on the types of goods and services⁶, and with whom Cambodians do business. As a result, the degree of dollarisation differs across regions, and concentrates in urban areas, tourism sites and near trade borders (Figure 3).

Since Cambodia was increasingly integrated into the global economy in the early 1990s, it has performed spectacularly (e.g., its average annual GDP growth is around 7%), but it has made it easy to use foreign currencies because products have been quoted in foreign currency. Hence, the degree of dollarisation has increased from 35% in 1993 to 84% in 2017. This upward trend of dollarisation in Cambodia appears to be the odd one out among other dollarised economies. To the best of our knowledge, the theoretical model we develop in the next section is the first effort to formally explain this counterfactual phenomenon.

4 MODEL

The theoretical framework is a cash-in-advance model (Uribe, 1997) that explicitly introduces the role of a foreign currency as a unit of account to amplify the network effects of using it. In addition, we incorporate a government's degree of regulations on the use of foreign currency in price quotation. These inclusions are motivated by observations in dollarised economies: Some dollarised economies, such as Laos P.D.R., Vietnam, and Peru, officially ban a foreign currency from being used as a unit of account, while Cambodia allows such a role of foreign currency in its economy.⁷

The economy is a small, open, dual-currency system in which a foreign currency, denoted the dollar, is used alongside a domestic currency, riel. This economy is endowed and populated by a large number of infinitely lived, identical households whose preferences are defined over paths of consumption, $\{C_t\}_{t=0}^{\infty}$, and represented by the following lifetime utility function:

$$\int_0^{\infty} e^{-\rho t} U(C_t) dt \tag{1}$$

⁶e.g., the US dollar is predominantly used to quote salaries and pay employees in the private sector as well as international organisations, and for big-ticket items, such as houses and vehicles.

⁷We have data of only these four countries about using a foreign currency as a unit of account. For other dollarised economies, we could not find any documents indicating whether they have allowed a foreign currency to be used in price quotations.

where ρ is a discount rate and $U(\cdot)$ is an instantaneous utility function which is assumed to be continuously differentiable, strictly increasing, and strictly concave. Consumption is a composite of a continuum of goods, $c_t(\theta)$, indexed by $\theta \in [0, 1]$,

$$C_t = \int_0^1 u(c_t(\theta)) d\theta,$$

where $u(\cdot)$ is assumed to be continuously differentiable, strictly increasing, and strictly concave, and to satisfy $\lim_{c \rightarrow 0} u'(c) = \infty$. The Inada condition implies that a consumption of every type of good θ is always positive: $c(\theta) > 0$.

Households can pay for goods in either riel or dollars. However, using dollars incurs a transaction cost that differs across goods θ . That is, the unit transaction cost of buying good θ with dollars is $\phi(\theta, k_t, q_t)$. The variable k_t represents the economy's accumulated experience at time t with using dollars as a medium of exchange, and it is referred to as *dollarisation capital*. The variable $q_t \in [0, 1]$ – referred to as *dollar pricing index* – is a proportion of products that merchants quote their prices in dollars.⁸ $q = 0$ means that all products' prices are displayed in the local currency, riel; $q = 1$ indicates all prices are quoted in the foreign currency, dollar. Unlike k_t , which is a demand-side stock of transaction experience, q_t is a firm-side state variable that measures the share of active price quotations denominated in dollars. While the paths of k_t and q_t are taken as given by individual households, both variables are endogenously determined and positively reinforce each other at the aggregate level. The function $\phi(\theta, k_t, q_t)$ satisfies the following assumption:

Assumption 1. $\phi : [0, 1] \times \mathbb{R}^+ \times [0, 1] \rightarrow \mathbb{R}^+$ is non-negative, twice continuously differentiable, and strictly convex in three arguments and satisfies $\phi_\theta > 0$, $\phi_k < 0$, $\phi_q < 0$, $\lim_{k \rightarrow \infty} \phi_k = 0$, $\lim_{q \rightarrow 1} \phi_q = 0$, and $\lim_{\theta \rightarrow 1} \phi(\theta, k, q) = \infty$.

Assumption 1 states that the transaction cost is strictly increasing in θ , and becomes infinitely large as θ approaches 1. This prohibitive cost ensures that there is always positive demand for the domestic currency, as observed in dollarised economies. But the transaction cost is strictly decreasing in both k_t and q_t . This decreased cost is used to capture the network externalities resulting from adopting the dollar as an alternative legal tender. In addition, the condition

$\lim_{k \rightarrow \infty} \phi_k = 0$ and $\lim_{q \rightarrow 1} \phi_q = 0$ is for technical convenience when characterising a steady state.

There are only two markets in this endowed economy: the financial market and the goods market. As in Lucas (1982), the financial market opens first, and the goods market opens only after the former closes. At time t , households hold a stock of riel-denominated wealth W_t . With this amount of money, they go to the financial market to obtain their desired balance of dollars

⁸Cambodia's survey data show that q was 28% in 2010 and 43% in 2014 for the economy as a whole.

(d_t) and riels (M_t) at the nominal exchange rate (E_t), expressed as riels per dollar. In addition, they can buy or sell an internationally traded, dollar-denominated bond, b_t , at a price of \$1 per unit. This bond bears the constant real interest rate $r > 0$ in dollars at time t . We assume strict purchasing power parity and perfect capital mobility to ensure that the real interest rate is the same in both the domestic and world economies. Denoting $w_t \equiv W_t/E_t$ and $m_t \equiv M_t/E_t$, the household's budget constraint in dollars is given by:

$$w_t = b_t + m_t + d_t \quad (2)$$

As the goods market opens, one member of the household goes to purchase the desired amount of each good θ , while another member receives an endowment of y_t units of each good θ . All goods θ are traded internationally at the common price of one dollar or E per unit. When selling their goods, producers can accept either riels or dollars. If they take riels, they have to wait until the financial market opens to convert riels into dollars. Denote Θ_t^d and Θ_t^m as the set of goods the household purchases with dollars and riels at time t , respectively. In continuous time, the household faces the following cash-in-advance constraints:

$$m_t \geq \int_{\Theta_t^m} c_t(\theta) d\theta \quad (3)$$

$$d_t \geq \int_{\Theta_t^d} c_t(\theta) d\theta \quad (4)$$

where $\pi_t \equiv \dot{E}_t/E_t$ is the depreciation rate (or inflation rate) at time t . We assume $r + \pi_t > 0$, so that holding domestic currency entails a positive opportunity cost, while $r > 0$ implies the same for dollar balances. Moreover, household does not have any debt after exiting the economy: $\lim_{t \rightarrow \infty} e^{-tr} w_t \geq 0$. Denoting $\tau_t \equiv T_t/E_t$ to be the government's transfers to households in terms of dollars, the household's evolution of real wealth is given by:

$$\dot{w}_t = r w_t - (r + \pi_t) m_t - r d_t - \int_{\Theta_t^m} c_t(\theta) d\theta - \int_{\Theta_t^d} [1 + \phi(\theta, k_t, q_t)] c_t(\theta) d\theta + y_t + \tau_t \quad (5)$$

The representative household chooses paths $\left[c_t(\theta)_{\theta \in (0,1)}, \Theta_t^m, \Theta_t^d, m_t, d_t \right]_{t=0}^{\infty}$ to maximise its lifetime utility Equation (1) subject to the constraints Equations (2)-(5) by taking as given the initial wealth (w_o) and paths $[\pi_t, q_t, k_t, y_t]_{t=0}^{\infty}$ and satisfying the No-Ponzi condition. Because both liquid assets are costly to hold, Equations (3) and (4) bind in equilibrium. The first-order conditions associated with the household's optimisation problem are therefore (hereafter, we drop time subscripts when no risk of confusion arises)

$$\Theta^d = [0, \bar{\theta}(k, q, \pi)) \quad (6)$$

$$\Theta^m = [\bar{\theta}(k, q, \pi), 1] \quad (7)$$

$$\bar{\theta}(k, q, \pi) = \begin{cases} 0 & \text{if } \phi(0, k, q) \geq \pi \\ \theta \text{ such that } \phi(\theta, k, q) = \pi & \text{otherwise} \end{cases} \quad (8)$$

$$c(\theta) = c(\bar{\theta}(k, q, \pi)) \quad \text{for } \theta \in \Theta^m \quad (9)$$

$$\frac{u_c(c(\theta))}{1 + r + \phi(\theta, k, q)} = \frac{u_c(c(\bar{\theta}(k, q, \pi)))}{1 + r + \pi} \quad \text{for } \theta \in \Theta^d \quad (10)$$

$$\dot{\lambda} = \lambda[\rho - r] \quad (11)$$

$$U_C(C)u_c(c(\theta)) = \lambda[1 + r + \phi(\theta, k, q)] \quad \text{for } \theta \in \Theta^d \quad (12)$$

Equations (6) and (7) show that there exists a cut-off good $\bar{\theta}(k, q, \pi) \in [0, 1]$ at time t , such that goods with index $\theta \geq \bar{\theta}(k, q, \pi)$ are bought with riels and the rest with dollars (Figure 4). This cut-off good is referred to as the degree of dollarisation at time t . Equation (8) shows that when the degree of dollarisation is positive, it is given by the good whose effective cost is the same whether it is purchased with dollars or with riels. Assumption 1 implies that $\bar{\theta}(k, q, \pi)$ is continuous and any triple (k, q, π) such that $\bar{\theta}(k, q, \pi) > 0$, $\bar{\theta}(k, q, \pi)$ is strictly increasing in all three arguments, continuously differentiable, and strictly concave in k and q (Figure 5). Equations (9) and (10) state that $c(\theta)$ is continuous and strictly decreasing in θ for $\theta \in \Theta^d$, and constant for $\theta \in \Theta^m$. Equation (11) implies that λ – the shadow price of wealth related to flow constraint Equation (5) – is constant over time because the interest rate equals the discount rate at the steady state. Equation (12) shows that the marginal utility of consumption of the goods bought with dollars is equal to the product of the shadow price of wealth and the effective price of the goods. In continuous time, this effective price equals the direct transaction cost of buying a good with dollars, $1 + \phi$, plus the opportunity cost of tying up dollar liquidity, r .

Following Uribe (1997), the stock of dollarisation capital evolves according to

$$\dot{k} = f(\bar{\theta}(k, q, \pi)) - \delta_k k \quad (13)$$

where $\dot{x}$ is the time derivative of x and δ_x refers to the depreciation rate of x . This law of motion captures social learning on the means-of-payment margin: as more goods are purchased with dollars, the economy accumulates more experience in using dollars in transactions.

Figure 4: Degree of dollarisation

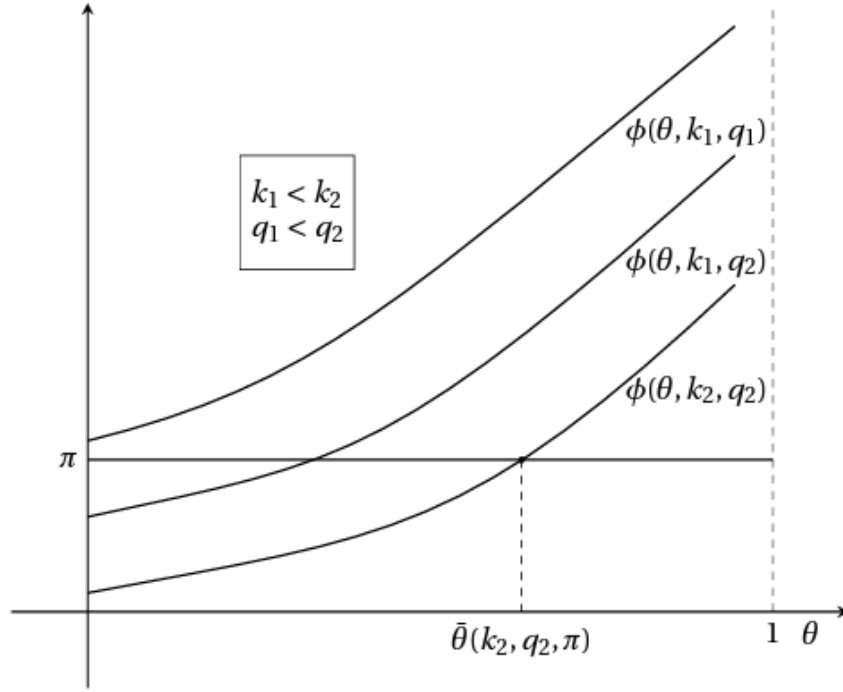
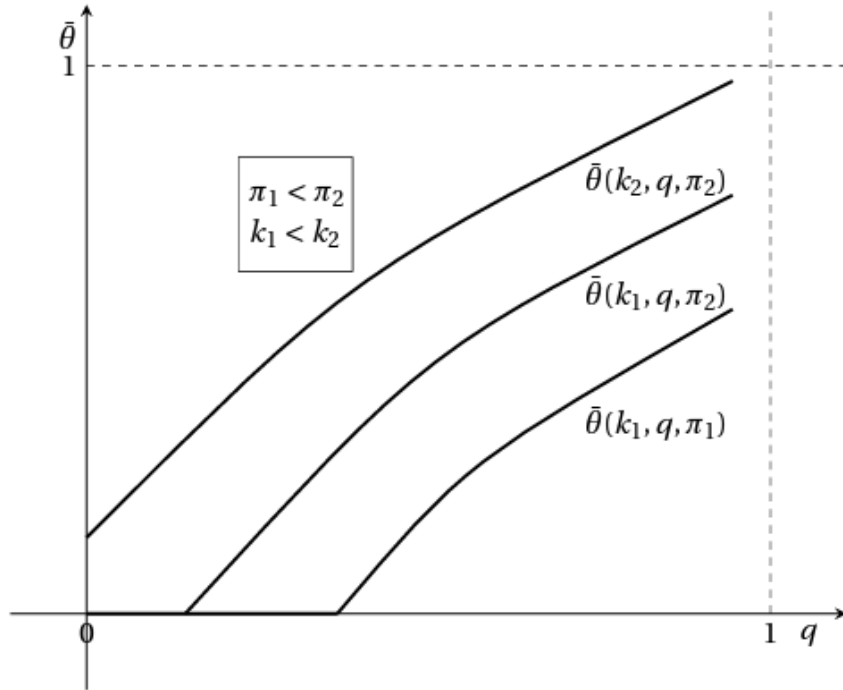


Figure 5: The function of $\bar{\theta}(k, q, \pi)$



The quotation decision and the law of motion of q

To model the unit-of-account margin explicitly, consider a continuum of firms of measure one. At each quotation opportunity, a firm decides whether to quote its price in riels or in dollars until the next opportunity. Let $\kappa_q > 0$ denote the Poisson arrival rate of quotation opportunities. A riel-quoting firm must update its posted local-currency price as the exchange rate changes, which entails a flow menu cost $\mu\pi_t$, where $\mu > 0$. A dollar-quoting firm avoids that menu cost, but when only a small fraction of goods is transacted in dollars, it incurs a flow customer-mismatch cost $\omega[1 - \bar{\theta}_t]$, where $\omega > 0$, because dollar quotation is less convenient when buyers and sellers are less accustomed to settling transactions in dollars. In addition, a dollar-quoting firm draws an idiosyncratic flow-equivalent maintenance cost ξ , distributed across firms according to the cumulative distribution function $G(\cdot)$ over the non-negative support.

Let V_t^m and $V_t^d(\xi)$ denote the present-discounted values of quoting in riels and dollars, respectively, conditional on receiving a quotation opportunity at date t . Normalising gross revenues to be the same under either quotation currency is innocuous here because, at a common effective price, the quotation currency changes the cost of maintaining the posted price rather than the underlying demand schedule. Hence, the quotation choice is driven by the discounted stream of quotation costs. In principle, the exact value comparison depends on the expected future paths of π and $\bar{\theta}$. To keep the aggregate system tractable, we adopt a local approximation in which firms treat aggregate conditions over the life of an individual quote as approximately constant at their date- t values. Under this approximation,

$$V_t^m = \frac{\bar{\Pi}_t - \mu\pi_t}{r + \kappa_q}, \quad V_t^d(\xi) = \frac{\bar{\Pi}_t - \omega[1 - \bar{\theta}_t] - \xi}{r + \kappa_q}.$$

Because the denominator is common and strictly positive, the present-value ranking is equivalent to the flow ranking, and a firm quotes in dollars whenever

$$\Delta_t(\xi) \equiv \mu\pi_t - \omega[1 - \bar{\theta}_t] - \xi \geq 0.$$

Hence, there exists a threshold maintenance cost

$$\bar{\xi}_t = \mu\pi_t - \omega[1 - \bar{\theta}_t],$$

and the inflow of newly posted prices quoted in dollars is

$$h(\bar{\theta}_t, \pi_t) \equiv \kappa_q G(\bar{\xi}_t).$$

Aggregating across firms yields the law of motion for the dollar pricing index:

$$\dot{q} = h(\bar{\theta}(k, q, \pi), \pi) - (\delta_q + \eta)q \quad (14)$$

where δ_q is the baseline attrition rate of existing dollar quotations and η refers to the degree of government regulations on using the dollar as a unit of account. In this formulation, regulation affects the survival of existing dollar quotations rather than the profitability ranking at the moment of posting, so it enters the outflow term of Equation (14). Appendix B reports the corresponding present-value comparison and aggregation argument. Equation (14) makes clear that q is distinct from k : k is a demand-side stock of experience in paying with dollars, while q is a supply-side stock of active dollar quotations chosen by firms.

Assumption 2. *$f : [0, 1] \rightarrow \mathbb{R}^+$ and $h : [0, 1] \times \mathbb{R}^+ \rightarrow \mathbb{R}^+$ are continuously differentiable and strictly increasing; f is strictly concave, while h is concave in its arguments and, for fixed inflation, strictly concave in its first argument whenever the density associated with G is strictly decreasing on the relevant interval, with $f(0) = 0$ and $h(0, 0) = 0$.*

Assumption 2 states that as more people use dollars to purchase goods, the amount of dollars circulating in the economy and the proportion of the economy quoting prices in dollars increase. Higher inflation also makes riel quotation more costly over the expected life of a price quote. Thus, there is social learning by doing on the payment side and an explicit quotation choice on the pricing side. The depreciation rates $\delta_k \in (0, 1]$ and $\delta_q \in (0, 1]$ together with $f(0) = 0$ and $h(0, 0)$ mean that the stock of dollarisation and that of the dollar pricing index gradually decrease as the dollar is no longer used in the economy – the economy as a whole forgets how to use the dollar and old dollar quotations disappear. The degree of regulations η captures the effects of government intervention in the use of the dollar in price quotation. The more stringent the regulations, the faster the fall of the dollar pricing index. For analytical convenience, η is assumed to be between 0 and 1 and satisfy $\eta + \delta_q \in (0, 1]$.

The Steady-State Equilibrium

To simplify and characterise the steady state, Equations (13) and (14) can be rewritten as

$$\dot{k} = F(k, q, \pi) - \delta_k k \quad (15)$$

$$\dot{q} = H(k, q, \pi) - (\delta_q + \eta)q \quad (16)$$

where F and H are the composites of functions f and θ , and h and (θ, π) , respectively. Assumption 2, together with the fact that θ is strictly increasing and concave in k and q , implies that F and H are also strictly increasing and concave in k and q .

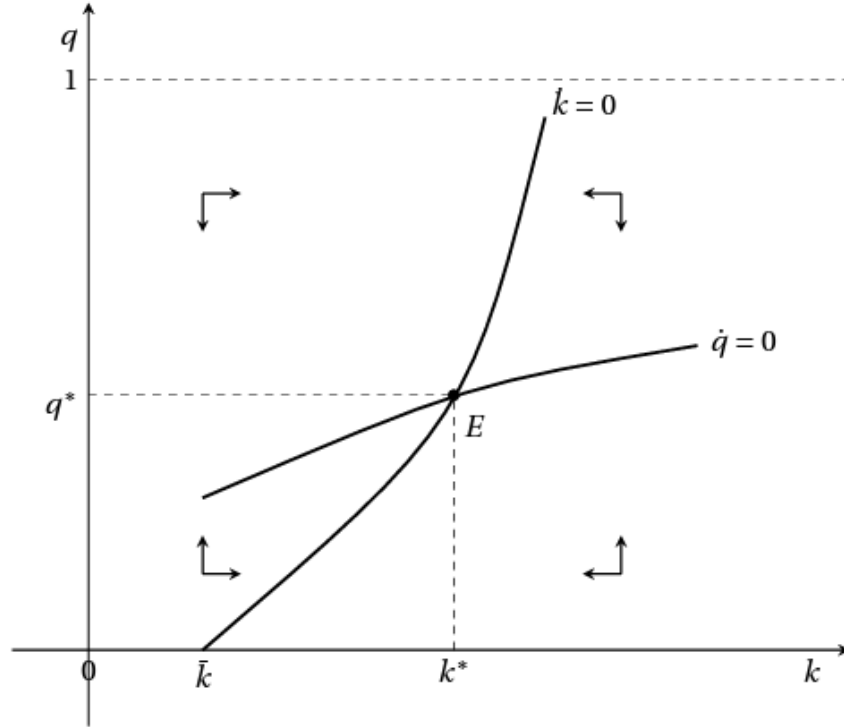
A steady-state equilibrium is defined by the stock of dollarisation and that of the dollar pricing index (k^*, q^*) , which satisfy the following two equations:

$$F(k^*, q^*, \pi) - \delta_k k^* = 0 \quad (17)$$

$$H(k^*, q^*, \pi) - (\delta_q + \eta)q^* = 0 \quad (18)$$

This steady-state equilibrium (k^*, q^*) is unique, and the phase diagram below indicates that it is stable. To see this heuristically, consider Figure 6, drawn in the (k, q) space. The curves $\dot{k} = 0$ and $\dot{q} = 0$, corresponding respectively to Equations (17) and (18), have a positive slope so that a higher dollar price index is associated with a higher dollarisation capital in equilibrium. Moreover, the slope of $\dot{k} = 0$ curve is steeper than that of $\dot{q} = 0$ curve at the equilibrium.

Figure 6: The dynamics of q and k



To show that these statements are true, first consider the slope of $\dot{k} = 0$ curve in the (k, q) space. Based on the implicit function theorem, this slope is given by:

$$\left. \frac{dq}{dk} \right|_{\dot{k}=0} = \frac{\delta_k - F_k(k, q, \pi)}{F_q(k, q, \pi)} \quad (19)$$

where $F_k \equiv \partial F / \partial k$ and $F_q \equiv \partial F / \partial q$. Equation (15) implies that $\delta_k = F(k, q, \pi) / k$ as $\dot{k} = 0$.

The fact that F is strictly concave in k and $F(0, q, \pi) \geq 0$ yields the following inequality:

$$\begin{aligned} F(k, q, \pi) &> F_k(k, q, \pi)k + F(0, q, \pi) \\ &> F_k(k, q, \pi)k \end{aligned}$$

This inequality leads to $\delta_k - F_k(k, q, \pi) > 0$, thus proving that Equation (19) is strictly positive. Similarly, when the same method and argument are applied to Equation (16), the slope of $\dot{q} = 0$ curve, given below, is also strictly positive:

$$\left. \frac{dq}{dk} \right|_{\dot{q}=0} = \frac{H_k(k, q, \pi)}{\delta_q + \eta - H_q(k, q, \pi)} \quad (20)$$

Second, to show that the slope of $\dot{k} = 0$ curve is steeper than that of $\dot{q} = 0$ curve at the equilibrium, we need to prove that the following inequality is true:

$$\begin{aligned} \left. \frac{dq}{dk} \right|_{\dot{k}=0} &> \left. \frac{dq}{dk} \right|_{\dot{q}=0} \\ &\Downarrow \\ \frac{F/k - F_k}{F_q} &> \frac{H_k}{H/q - H_q} \\ &\Downarrow \\ (F - kF_k)(H - qH_q) &- qkF_qH_k > 0 \\ &\Downarrow \\ kH_k(F - kF_k - qF_q) + qF_q(H - kH_k - qH_q) &+ (F - kF_k - qF_q)(H - kH_k - qH_q) > 0 \end{aligned}$$

The second line is a result of substituting $F(k^*, q^*, \pi)/k^*$ for δ_k in Equation (19) and $H(k^*, q^*, \pi)/q^*$ for $\delta_q + \eta$ in Equation (20). The third and last lines result from simplifying and rearranging the terms in the second line (note that all functions are evaluated at the equilibrium point). Assumptions 1 and 2 together with Equation (8) show that this inequality is satisfied (see proof in Appendix B).

Third, the dynamic movements of k and q are dictated by the signs of the time derivatives of k and q , respectively. Since $\partial \dot{k} / \partial q = F_q > 0$, all the points below the $\dot{k} = 0$ curve are characterised by $\dot{k} < 0$ and all the points above the curve by $\dot{k} > 0$. Similarly, because $\partial \dot{q} / \partial k = H_k > 0$, all the points to the left of the $\dot{q} = 0$ curve are characterised by $\dot{q} < 0$ and all the points to the right of the curve by $\dot{q} > 0$. Therefore, the $\dot{k} = 0$ curve and $\dot{q} = 0$ curve divide the phase space into four regions, each with its own distinct pairing of signs of $\dot{k}$ and $\dot{q}$, as indicated by the right-angled directional arrows in Figure 6.

Finally, since this paper focuses on developing economies that have already experienced high inflation, the steady-state stock of dollarisation in the Uribe (1997) benchmark is positive while the dollar pricing index is 0 ($q = 0$). This steady-state stock is denoted by $\bar{k}$ in Figure 6. At the point $(\bar{k}, 0, \pi)$, the $\dot{q} = 0$ curve lies above the $\dot{k} = 0$ curve because $\dot{q} = H(\bar{k}, 0, \pi)$ is greater than 0 ($\bar{\theta}(\bar{k}, 0, \pi) > 0$). Assumption 1 implies that as q approaches to 1, the slope of the $\dot{k} = 0$ curve approaches to infinity. Likewise, as k approaches to infinity, the slope of $\dot{q} = 0$ approaches to 0.

Therefore, the two demarcation curves ($\dot{k} = 0$ and $\dot{q} = 0$) intersect only once at point $E(k^*, q^*)$. In the relevant region of the phase diagram, this point is the unique interior steady state, and the directional arrows indicate convergence toward point E from the surrounding regions of the phase space. We do not attempt a separate formal global-stability proof.

Uribe (1997) as a benchmark

The present framework nests the Uribe (1997) environment when the quotation margin is shut down. If all prices are quoted in the domestic currency so that $q_t \equiv 0$, the transaction cost becomes $\phi(\theta, k_t, 0)$ and the dynamic system collapses to

$$\dot{k} = F(k, 0, \pi) - \delta_k k.$$

This is a Uribe-style one-state benchmark within the present functional environment, rather than a claim to reproduce all features of Uribe's original specification, including its multiple-equilibrium structure. In that benchmark, the economy's adjustment is driven only by the interaction between inflation and dollarisation capital. The regulation parameter η plays no independent role because there is no separate unit-of-account margin to regulate.

By contrast, once firms are allowed to choose the currency of quotation, regulation affects the economy through two channels. First, a change in η directly changes the survival of dollar quotations. Second, because $\phi_q < 0$, a lower dollar pricing index raises the transaction cost of using dollars and feeds back into the accumulation of dollarisation capital. This additional feedback is the main methodological difference from Uribe (1997), and it is what allows the model to distinguish Cambodia – where the quotation margin is effectively unrestricted – from economies such as Laos P.D.R., Vietnam, and Peru, where restrictions on foreign-currency quotation weaken both the unit-of-account and means-of-payment roles of the dollar.

Why the extension matters relative to Uribe (1997)

The benchmark comparison can be stated in three steps. First, when inflation falls, both models predict weaker incentives to transact in dollars; however, only the present framework predicts an additional reduction in dollar quotation because lower inflation also lowers the menu-cost saving from posting prices in dollars. Second, there is no counterpart to Figure 8 in the

Uribe benchmark because quotation policy η is absent when $q = 0$. Third, the joint experiment in Figure 9 – in which lower inflation works against dollarisation while a more permissive quotation policy works in its favour – is also absent from the benchmark. For that reason, the Uribe model is a useful benchmark for persistence, but the present model is needed to analyse economies in which policy toward foreign-currency quotation differs across countries.

Another implication of the extension is that the present specification is analysed around a unique interior steady state in the relevant region of the phase diagram, rather than around the multiple-equilibrium structure emphasised by Uribe (1997). The gain from that restriction is that the quotation-policy margin can be studied transparently; the cost is that the richer hysteresis structure of the benchmark is not preserved in the current formulation.

5 THE EFFECTS OF GOVERNMENT POLICIES

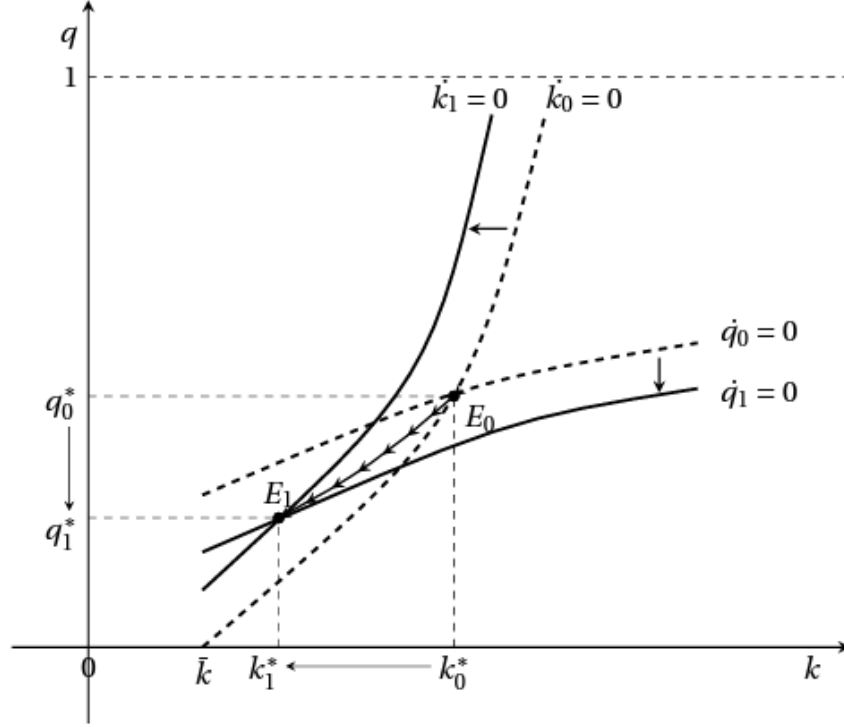
In this section, we illustrate how changes in government policies affect the steady-state equilibrium and present three relevant cases to explain the rise and fall of the degree of dollarisation in developing countries.

5.1 THE EFFECTS OF CHANGES IN INFLATION

First, consider what happens to the equilibrium when a government reduces inflation. Empirically, when high-inflation countries reduced inflation to a low level for a period of time, their dollarisation has gradually declined (see, e.g., Figure 10 and Figure A1). Figure 7 explains how this mechanism works.

In the structural model developed above, lower inflation reduces not only the incentive to hold dollars for transactions but also the menu-cost advantage of quoting prices in dollars.

Figure 7: The effects of falling inflation

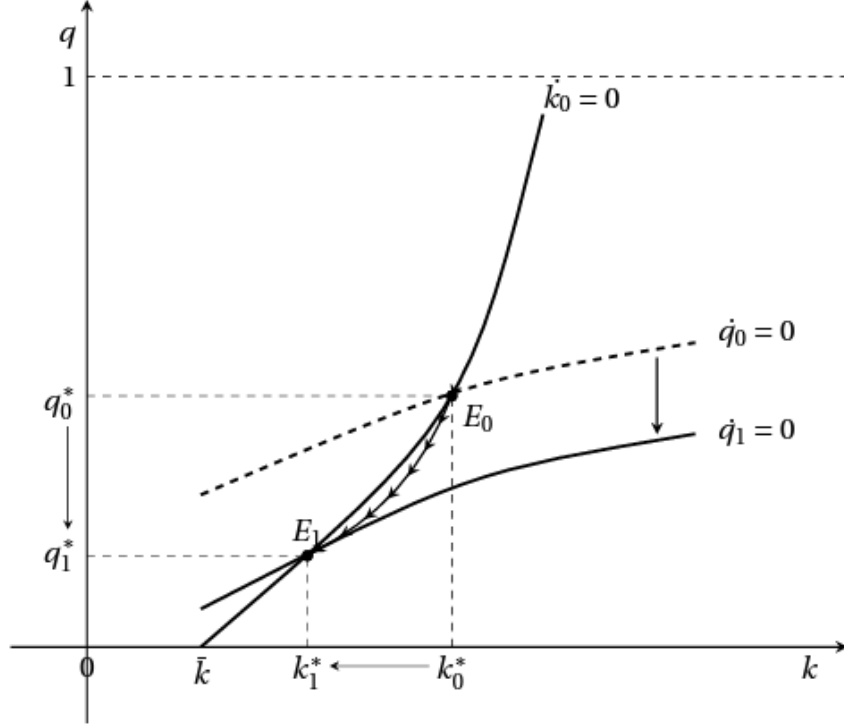


A decrease in inflation causes the $\dot{k} = 0$ curve to shift to the left, from the $\dot{k}_0 = 0$ curve to the $\dot{k}_1 = 0$ curve; but the $\dot{q} = 0$ curve moves downward, from the $\dot{q}_0 = 0$ curve to the $\dot{q}_1 = 0$ curve. These directional movements are derived from the fact that

$$\left. \frac{dq}{d\pi} \right|_{\dot{k}=0} = -\frac{F_\pi(k, q, \pi)}{F_q(k, q, \pi)} < 0, \text{ and } \left. \frac{dq}{d\pi} \right|_{\dot{q}=0} = \frac{H_\pi(k, q, \pi)}{\delta_q + \eta - H_q(k, q, \pi)} > 0$$

Now the economy is governed by the new curves $\dot{k}_1 = 0$ and $\dot{q}_1 = 0$. At the point E_0 , both q_0^* and k_0^* are at such a high level that they cannot generate enough social learning by adopting the foreign currency to compensate for their depreciation. Hence, both stocks gradually decline along the transition path E_0E_1 until they reach a new steady state at point E_1 , where the new level of both stocks are lower than before.

Figure 8: The effects of increasing regulation on dollar quotation



5.2 THE EFFECTS OF CHANGES IN DOLLAR QUOTATION

Next, consider how the equilibrium changes when a government imposes restrictions on the quotation of the dollar. In the structural quotation block, such restrictions increase the attrition of existing dollar quotations. Restricting the use of the dollar as a unit of account, therefore, affects only the $\dot{q} = 0$ curve, making the stock of the dollar pricing index depreciate more quickly. In other words, if the other variables remain the same, an increase in η – the degree of regulations on the use of the foreign currency as price quotation – leads to a decrease in q . That is,

$$\left. \frac{dq}{d\eta} \right|_{\dot{q}=0} = -\frac{q}{\delta_q + \eta - H_q(k, q, \pi)} < 0$$

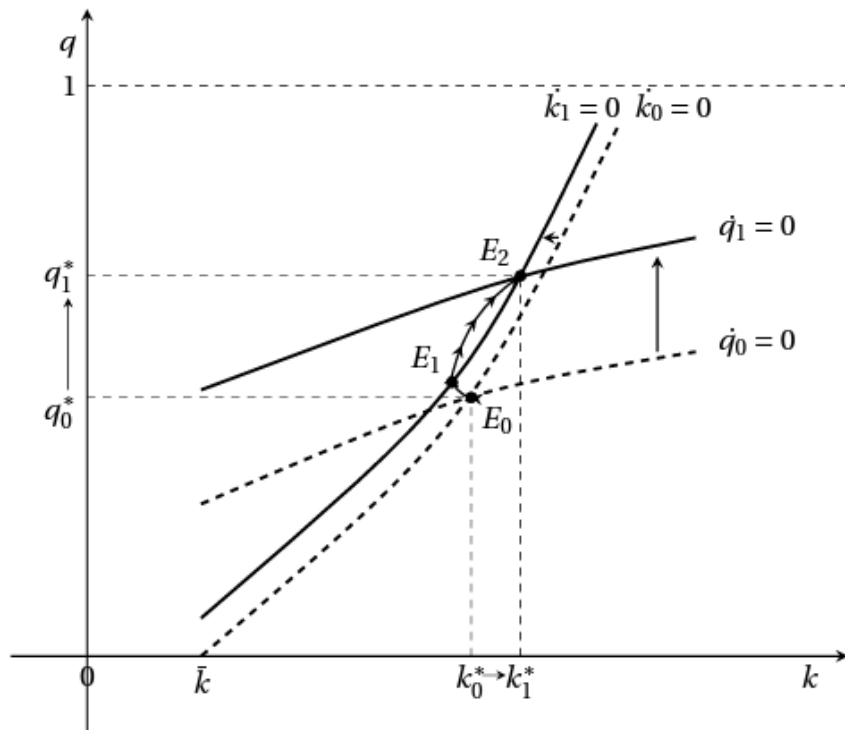
As the curve $\dot{q} = 0$ shifts downward, from the $\dot{q}_0 = 0$ curve to the $\dot{q}_1 = 0$ curve (Figure 8), the equilibrium moves along the transition path $E_0 E_1$ from the point E_0 to the point E_1 , where the stock of dollarisation and dollar pricing index capitals are lower than before.

5.3 THE EFFECTS OF CHANGES IN INFLATION AND DOLLAR QUOTATION

Finally, consider a case in which a government has lowered inflation but has relaxed restrictions on using the dollar as a unit of account. In this case, the household payment margin works

against dollarisation, but the firm quotation margin works in its favour. Melvin and Fenske (1992), Clements and Schwartz (1993), and Valev (2010) provide evidence that network externalities are stronger than currency risk. This evidence implies that the shift of the $\dot{q} = 0$ curve (from the $\dot{q}_0 = 0$ to the $\dot{q}_1 = 0$), as shown in Figure 9, is larger than that of the $\dot{k} = 0$ curve (from the $\dot{k}_0 = 0$ to $\dot{k}_1 = 0$). Since the point E_0 is below both new curves, the stock of dollarisation initially declines, but that of dollar pricing index increases along the transition path $E_0E_1E_2$ from E_0 to E_1 . From E_1 onward, both stocks gradually rise to reach the new equilibrium at the point E_2 , where the level of both stocks are higher than before.

Figure 9: The effects of a decrease in inflation and an increase in dollar quotation



6 THE MODEL PREDICTIONS AND EVIDENCE

The empirical discussion uses three related but distinct measures of dollarisation. Table 1 summarises these objects and their roles in the paper. In particular, the dollar pricing index is not inferred mechanically from deposit dollarisation; it is measured separately from Cambodian survey evidence and serves as the empirical counterpart to the unit-of-account margin in the model.

Table 1: Three empirical measures used in the paper

Measure	Empirical proxy	Role in the paper
Deposit dollarisation	Ratio of foreign-currency deposits to broad money (M2)	Cross-country time-series comparison in Figures 1, 10, and A1
Transaction dollarisation	Share of revenues or transactions received in foreign currency	Empirical counterpart to the means-of-payment margin in Cambodia regions
Dollar pricing index (q)	Share of goods whose prices are quoted in foreign currency	Survey-based empirical counterpart to the unit-of-account margin

Note: The dollar pricing index is measured from Cambodian survey data and is therefore distinct from both deposit dollarisation and transaction dollarisation.

The illustrations in Figures 7 to 9 suggest the explanations behind the upward trend of dollarisation in Cambodia and the downward trends in Laos, Vietnam, and Peru. This comparison is also where the extension relative to Uribe (1997) becomes essential. In the Uribe benchmark, these economies could differ only because of different inflation histories. In the present model, they may also differ because governments adopt different regimes toward foreign-currency quotation, captured by η . Since its first national election in 1993, Cambodia has maintained an average depreciation rate of the riel around 2%. However, the Cambodian government has acquiesced to the use of the dollar and has imposed no restrictions on the use of foreign currencies. In particular, the government has not imposed any regulation on which currency is used to quote products' prices (Kubo, 2017). This implies that η – the degree of regulations on using the dollar for price quotation – approaches 0. Because the network externalities of using the dollar are stronger than the currency risk, the degree of dollarisation has increased (Table 2 and Figure 10).

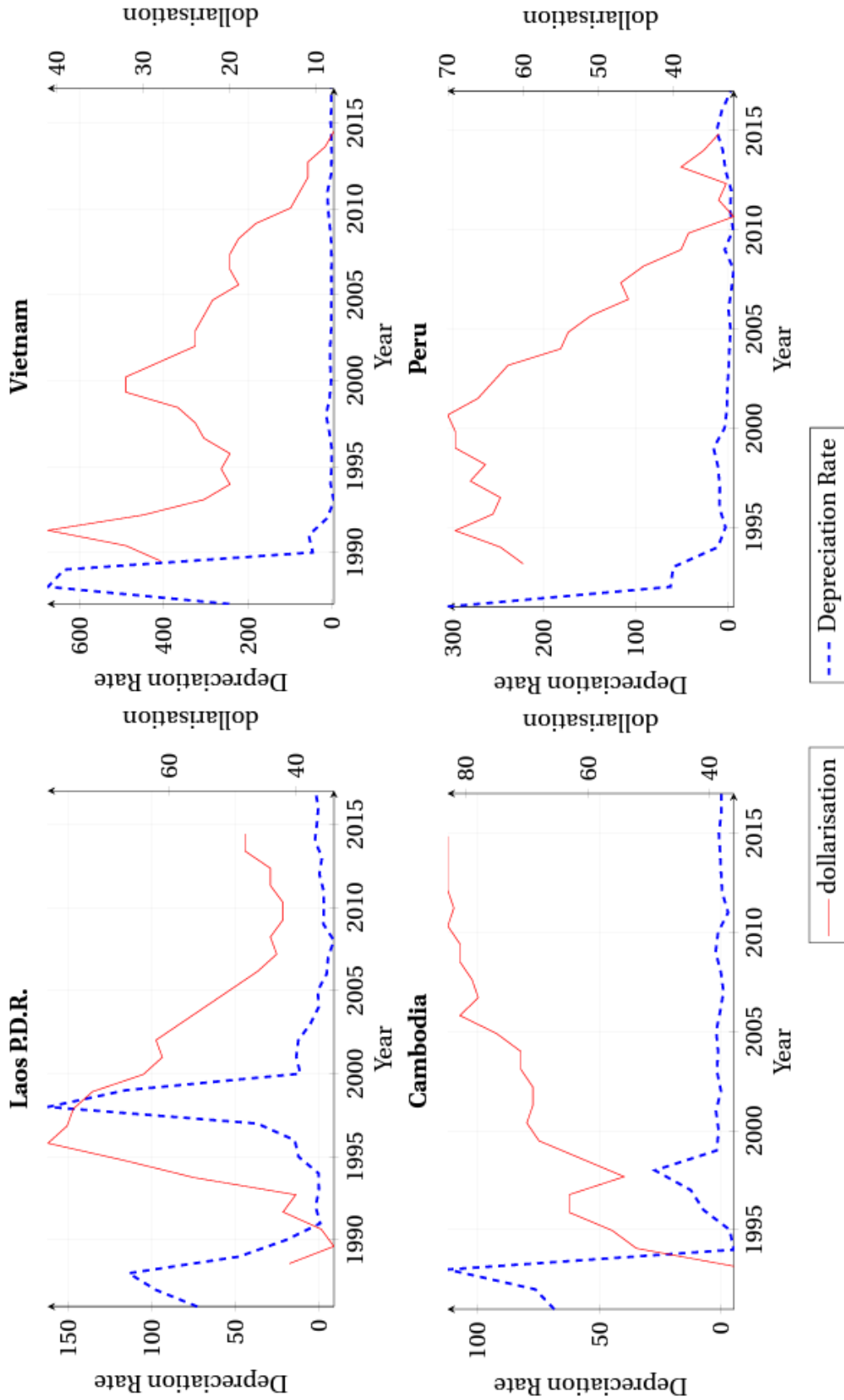
Table 2: Regulations on dollarisation in four selected countries

	Quotation of Prices in Foreign Currency
Cambodia	No control
Laos P.D.R.	In 1990, Laos issued Decree No53/CM to prohibit the use of foreign currency for domestic transactions, but it was not strictly enforced. In March 2008, the decree was upgraded for stringent enforcement with penalties
Vietnam	Vietnam banned (but did not strictly enforce) foreign currency for quoting prices in the 1990s. In 2011, the restriction was upgraded to raise the penalties to a maximum of VND 500million.
Peru	In 2004, Peru enacted a law stipulating that all prices be denominated in soles, the local currency.

Note: Data for Cambodia, Vietnam, and Laos P.D.R. are sourced from Kubo (2017), and data for Peru are obtained from Castellares et al. (2019).

In contrast, Laos P.D.R., Vietnam, and Peru have repeatedly restricted their economies by forbidding price quotations in foreign currency. For instance, as in Cambodia, it has been a common practice in Laos P.D.R. for firms and households to use foreign currencies as a means of exchange and unit of account instead of its local currency, the kip. The Thai baht and the U.S. dollar are widely used throughout the country, particularly in urban areas, tourist destinations, and at Thailand’s borders. However, in the 2000s, the Laotian government imposed regulations on dollar-denominated bank lending and prohibited pricing in dollars. To enforce the regulations, the Bank of the Laos P.D.R. even established a committee to conduct regular inspections to monitor the price quotation of products in markets, shops, companies, and trade fairs or exhibitions (Kubo, 2017). This regulation has increased the cost of using the foreign currency. The higher costs and lower depreciation rates in Laos P.D.R., Vietnam, and Peru have reduced the degree of dollarisation in these economies (Table 2 and Figure 10). This quotation-policy channel is precisely the margin that is absent from the Uribe benchmark.

Figure 10: Dollarisation trends in Cambodia, Laos P.D.R., Vietnam, and Peru



Note: The degree of dollarisation is a ratio of foreign currency deposits to broad money (M2). The data to construct dollarisation are sourced from various IMF Country Reports (Appendix A) and the exchange rate is obtained from International Financial Statistics (2019).

A disciplined minimum-distance benchmark against Uribe (1997)

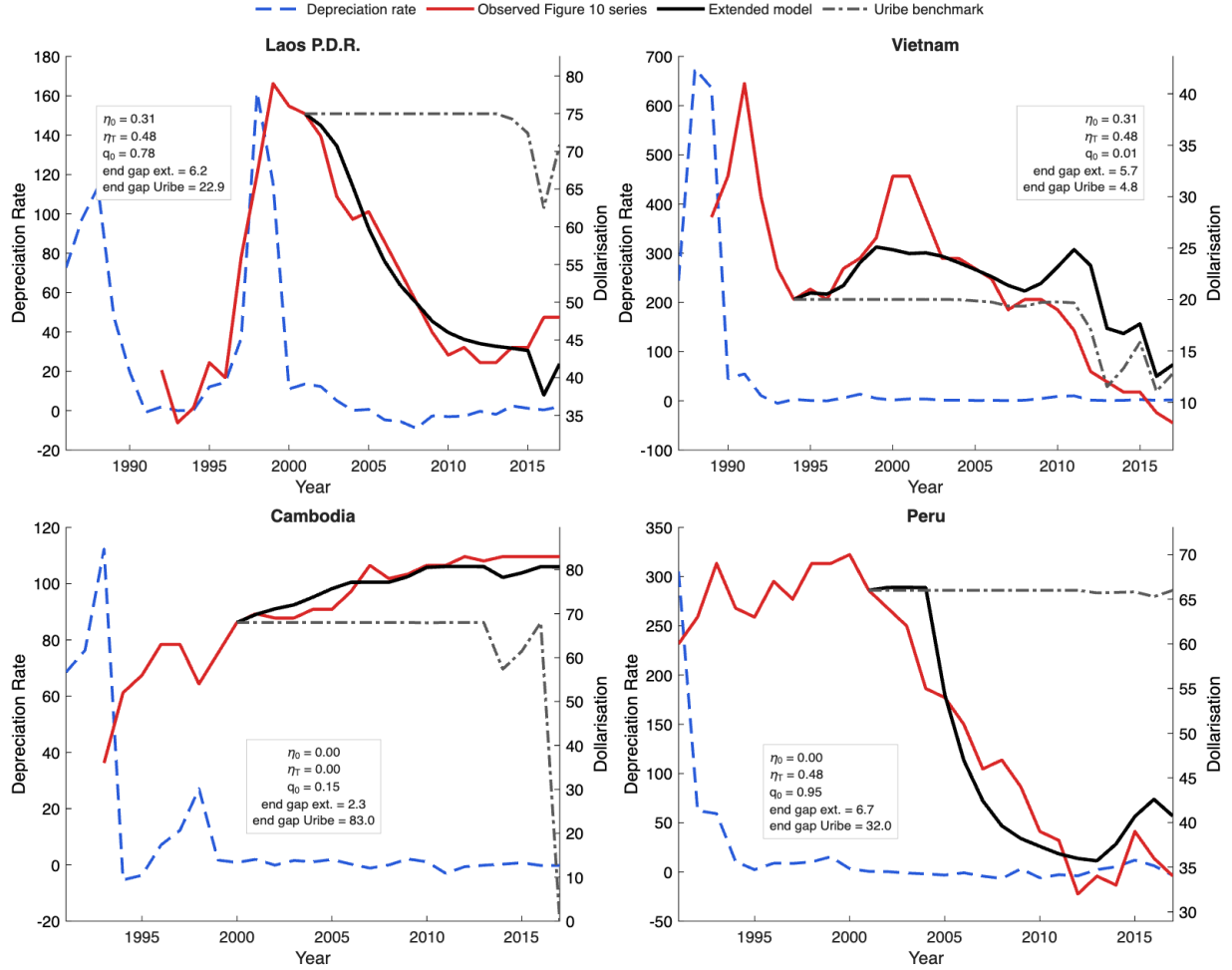
Figure 10 and Table 2 are descriptive. To assess more directly whether the quotation margin is empirically important relative to the Uribe benchmark, we estimate a simple minimum-distance transition exercise for the same four countries. For each country, the post-stabilisation depreciation path is fed into both models under a common structural parameterisation, and the simulated paths are compared with the original deposit-dollarisation series used to build Figure 10. The common parameters, two policy-intensity levels (η^W, η^S), and the inherited quotation stocks q_0 are chosen to minimise squared deviations between the simulated and observed deposit-dollarisation paths over the benchmark windows, together with squared deviations from the two Cambodian quotation-share moments reported in the paper (28% in 2010 and 43% in 2014). The policy wedge itself is not free by country: it follows the dated regulatory schedule implied by Table 2, with Cambodia at $\eta_t = 0$, Laos P.D.R. moving from η^W to η^S in 2008, Vietnam moving from η^W to η^S in 2011, and Peru moving from 0 to η^S in 2004. Because Figure 10 reports deposit dollarisation rather than the transaction cut-off $\bar{\theta}$ in the model, the benchmark is primarily disciplined by transition dynamics rather than unconditional levels. Keeping deposit dollarisation as the common cross-country outcome also avoids mixing different empirical measures across countries. In the extended model, the benchmark observable is the broad-dollarisation composite $D_t = \nu_D q_t + (1 - \nu_D)\bar{\theta}_t$, with $\nu_D = 0.48$, whereas in the Uribe benchmark it is $D_t^U = \bar{\theta}_t$. This reduced-form bridge is motivated by evidence that domestic price and asset dollarisation are positively related across countries (Drenik and Perez, 2021) and by Peruvian evidence that local-currency pricing regulation is associated with lower financial dollarisation (Castellares and Toma, 2019). Figure 11 reproduces the full observed Figure 10 series and overlays the simulated benchmark paths from the post-stabilisation start year onward after anchoring each model path to the first observed dollarisation level in that benchmark sample. Appendix C states the benchmark functional forms and the policy mapping used in this numerical exercise, while Appendix D reports robustness to alternative measurement-bridge weights.

Table 3: Minimum-distance benchmark estimates and terminal levels

Country	Sample	Mean dep.	η_0	η_T	q_0	Obs. end	Ext. end	Uribe end
Cambodia	2000–2017	0.35	0.00	0.00	0.15	83.0	80.7	0.0
Laos P.D.R.	2001–2017	0.49	0.31	0.48	0.78	48.0	41.8	70.9
Vietnam	1994–2017	3.20	0.31	0.48	0.01	8.0	13.7	12.8
Peru	2001–2017	-0.29	0.00	0.48	0.95	34.0	40.7	66.0

Note: “Mean dep.” is the average annual depreciation rate over the post-stabilisation sample used in the benchmark. “Obs. end”, “Ext. end”, and “Uribe end” report end-of-sample dollarisation levels; the two model paths are anchored to the first observed level in each country sample so that Figure 11 remains on the same scale as Figure 10. The benchmark estimates common structural parameters and two policy-intensity levels, $\eta^W = 0.3138$ and $\eta^S = 0.4781$, using minimum-distance estimation. The policy schedule is imposed from Table 2: Cambodia keeps $\eta_t = 0$, Laos P.D.R. moves from η^W to η^S in 2008, Vietnam moves from η^W to η^S in 2011, and Peru moves from 0 to η^S in 2004. Common structural parameters are $A_\phi = 0.1391$, $a_k = 1.0315$, $a_q = 2.2442$, $a_f = 0.0996$, $\alpha_f = 0.6762$, $\delta_k = 0.1154$, $\mu = 2.0267$, $\omega = 0.7920$, $\gamma = 4.3832$, $\kappa_q = 0.2937$, $\delta_q = 0.0005$, $\nu_D = 0.4797$, and $k_0 = 24.91$, where ν_D is the weight on q_t in the measurement bridge $D_t = \nu_D q_t + (1 - \nu_D)\bar{\theta}_t$. The two Cambodian quotation-share moments (28% in 2010 and 43% in 2014) also enter the minimum-distance objective. Appendix C reports the associated functional forms.

Figure 11: Minimum-distance benchmark against Uribe (1997)



Note: The observed dollarisation and depreciation series are the full original series used to construct Figure 10. To preserve comparability with Figure 10, the depreciation path is plotted on the left axis, and the dollarisation series on the right. The two model paths are overlaid from the benchmark start year onward and are anchored to the first observed dollarisation level in that benchmark sample. The extended model uses the common minimum-distance parameterisation in Table 3, the dated policy schedules implied by Table 2, and country-specific initial quotation stocks q_0 . The Uribe benchmark sets $q_t \equiv 0$.

The disciplined benchmark makes three points. First, in Cambodia, the Uribe benchmark still predicts a sharp fall in dollarisation once inflation is low, whereas the extended model keeps the series close to its initial level and comes much closer to the observed upward drift. Second, in Laos P.D.R. and Peru, where quotation restrictions tighten over time, the extended model generates substantially stronger de-dollarisation than the benchmark and is much closer to the data. Third, Vietnam remains the hardest case: the extended model improves only modestly on Uribe there and still overpredicts the end-of-sample level. The benchmark also disciplines the

latent quotation margin directly through the two Cambodian survey moments, which the model matches only approximately (37.5 versus 28 in 2010 and 37.9 versus 43 in 2014). Appendix D shows that these conclusions are not knife-edge with respect to the measurement bridge: re-estimating the benchmark with alternative weights on q_t in the observable leaves the extended model better than Uribe in all four countries. Accordingly, Figure 11 should be read as a disciplined minimum-distance benchmark rather than as a full structural estimation. Its purpose is to show that the quotation margin is empirically relevant.

7 CONCLUDING REMARKS

In this paper, we provide the first explanation for the puzzle of dollarisation in Cambodia. To this end, we develop a cash-in-advance model by introducing a new stock variable, the *dollar pricing index*, and a degree of government regulation over that variable. This index, defined as the share of active prices quoted in the foreign currency, is derived from an explicit firm-side quotation decision and amplifies the network effects of dollar usage. Because the impacts of network externalities have proven more substantial than those of currency risk, the positive interaction between the dollar's use as a unit of account and as a medium of exchange increases its use, even though Cambodia has achieved economic and political stability. The disciplined minimum-distance benchmark against Uribe (1997) clarifies that the contribution of the paper is not to replace hysteresis, but to add a quotation-policy channel that allows otherwise similar low-inflation economies to display different dollarisation paths; the Cambodia, Laos P.D.R., and Peru comparisons are particularly informative, while Vietnam remains a harder quantitative case.

Our model has important implications for dollarised economies. To reduce the widespread use of a foreign currency, dollarised economies need to stabilise not only their macroeconomic conditions (e.g. reduce inflation) but also increase the transaction cost of the use of the foreign currency (significantly restricting the foreign currency from being used as a unit of account because foreign currency-denominated goods explicitly promote the use of the foreign currency).

Our model has one limitation, though: it produces only one stable steady state in the relevant region of the phase diagram; hence, it cannot capture the richer hysteresis patterns that arise in the Uribe (1997) benchmark. However, since our work focuses on developing countries that have already experienced high inflation and thus a certain degree of dollarisation, our model suffices to deal with the present objective: explain whether this particular degree of dollarisation continues to rise or fall as a result of the government's regulations on the use of a foreign currency as a unit of account.

Declaration of conflict of interest: The authors declare no conflict of interest.

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APPENDIX

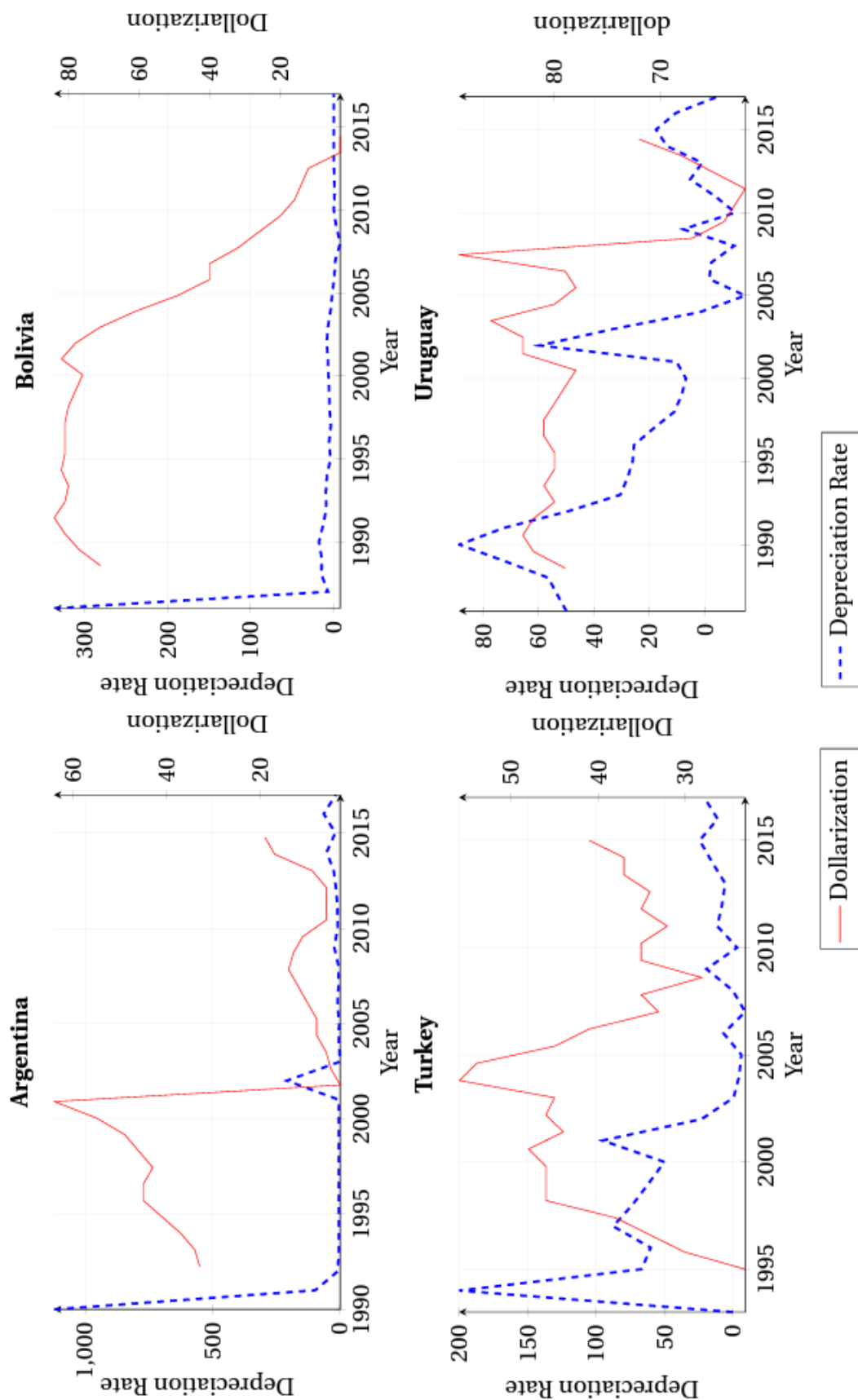
A Data Appendix and Supplementary Figure

Table A1: Data sources used to construct dollarisation in eight selected countries

	Sources
Cambodia	IMF Staff Country Reports: No. 95/108, No.99/33, No.00/134, No.06/265, No.09/325, No.11/45, No.14/33, and No.18/369.
Laos P.D.R.	IMF Staff Country Reports: No.96/54, No.98/77, No.3, No.02/62, No.04/394, No.06/398, No.08/340, No.11/44, No.13/369, No.15/45, and No.17/53.
Vietnam	IMF Staff Country Reports: No.95/93, No.99/56, No.03/382, No.07/386, No.12/165, No.16/240, and No.18/215.
Peru	IMF Staff Country Reports: No.95/108 , No.99/33 , No.00/160, No.06/265, No.09/325, No.11/45, No.14/33, and No.18/369.
Argentina	IMF Staff Country Reports: No.95/110, No.98/38, No.00/134, No.03/226, No.05/236, and No.17/409.
Bolivia	IMF Staff Country Reports: No.95/24, No.97/99, No.00/38, No.03/258, No.05/393, No.07/248, No.10/27, No.12/149, No.15/334, and No.18/379.
Turkey	IMF Staff Country Reports: No.96/112, No.00/14, No.02/138, No.05/163, No.07/362, No.10/278, No.13/363, No.16/104, and No.18/110.
Uruguay	IMF Staff Country Reports: No.95/75, No.99/102, No.01/183, No.04/327, No.06/425, No.08/45, No.10/43, No.14/6, No.15/81, and No.18/23.

Note: Dollarisation is a ratio of foreign currency deposits to broad money (M2). Data for Cambodia are supplemented by the data from the Ministry of Economy and Finance, Cambodia, where the author Veasna Kheng worked as an economist in 2015.

Figure A1: Dollarisation and exchange-rate movements in selected highly dollarised economies



Note: The degree of dollarisation is a ratio of foreign currency deposits to broad money (M2). The data to construct dollarisation are sourced from various IMF Country Reports and the exchange rate is obtained from International Financial Statistics (2019).

B Model Appendix and Proofs

B.1 A structural derivation of the dollar pricing index

Consider a unit mass of firms that receive quotation opportunities at Poisson rate $\kappa_q > 0$. Conditional on receiving such an opportunity at date t , a firm chooses its quotation currency until the next opportunity arrives. Normalise the common revenue component of a newly posted price to $\bar{\Pi}_t$; as in the main text, this normalisation is innocuous because at a common effective price the quotation currency affects the cost of maintaining the posted price rather than the underlying demand schedule. The flow payoff from quoting in riels is

$$\Pi_t^m = \bar{\Pi}_t - \mu\pi_t,$$

where $\mu\pi_t$ is the flow menu cost of keeping a local-currency quotation aligned with the exchange rate. The flow payoff from quoting in dollars after drawing firm-specific maintenance cost ξ is

$$\Pi_t^d(\xi) = \bar{\Pi}_t - \omega[1 - \bar{\theta}_t] - \xi.$$

Let V_t^m and $V_t^d(\xi)$ denote the present-discounted values of these two quotation choices. The exact value comparison is

$$\begin{aligned} V_t^m &= E_t \int_0^\infty e^{-(r+\kappa_q)s} [\bar{\Pi}_{t+s} - \mu\pi_{t+s}] ds, \\ V_t^d(\xi) &= E_t \int_0^\infty e^{-(r+\kappa_q)s} [\bar{\Pi}_{t+s} - \omega[1 - \bar{\theta}_{t+s}] - \xi] ds. \end{aligned}$$

The discount factor includes both time discounting and the hazard κ_q that the current quotation spell ends. Under the local approximation used in the main text, firms treat aggregate conditions over the life of an individual quote as approximately constant at their date- t values. Therefore,

$$V_t^m = \frac{\bar{\Pi}_t - \mu\pi_t}{r + \kappa_q}, \quad V_t^d(\xi) = \frac{\bar{\Pi}_t - \omega[1 - \bar{\theta}_t] - \xi}{r + \kappa_q}.$$

The denominator is common and strictly positive, so a firm chooses dollar quotation if and only if

$$\xi \leq \bar{\xi}_t \equiv \mu\pi_t - \omega[1 - \bar{\theta}_t].$$

Without the local approximation, this threshold would depend on the expected future path of π and $\bar{\theta}$ rather than just their date- t values. Because the support of ξ is non-negative, negative values of $\bar{\xi}_t$ imply no new dollar quotation. Otherwise, the mass of newly posted prices quoted

in dollars per unit of time is

$$h(\bar{\theta}_t, \pi_t) = \kappa_q \int_0^{\bar{\xi}_t} dG(\xi) = \kappa_q G(\bar{\xi}_t).$$

Since $\bar{\xi}_t$ is increasing in both $\bar{\theta}_t$ and π_t , the function h is increasing in those arguments; if the density associated with G is weakly decreasing on the relevant region, then h is concave there, and for fixed π_t it is strictly concave in $\bar{\theta}_t$. These properties match the monotonicity and concavity conditions imposed in Assumption 2. If active dollar quotations disappear at baseline rate δ_q because price lists, menus, and informal conventions are refreshed, and at the additional rate η because regulation forces a return to riel quotation, then the stock of dollar quotations evolves according to

$$\dot{q}_t = h(\bar{\theta}_t, \pi_t) - (\delta_q + \eta)q_t,$$

which is Equation (14) in the main text.

B.2 Proof of the slope-ordering inequality

The following proof establishes the inequality displayed after Equation (20). First, rewrite the expression $(F - F_k k)(H - H_q q) - F_q H_k q k$ as:

$$(F - F_k k - F_q q)(H - H_k k - H_q q) + H_k k(F - F_k k - F_q q) + F_q q(H - H_k k - H_q q)$$

Next, show that the terms in parentheses are positive. Assumption 1 implies that

$$\phi_\theta[\theta_2 - \theta_1] + \phi_k[k_2 - k_1] + \phi_q[q_2 - q_1] > \phi(\theta_2, k_2, q_2) - \phi(\theta_1, k_1, q_1); \forall \pi \quad (\text{B.1})$$

where $k_2 > k_1$ and $q_2 > q_1$ and thus $\theta_2 > \theta_1$ (θ strictly increases in k and q). Adding and subtracting π from the right-hand side of inequality (B.1) and applying Equation (8), we obtain:

$$[\phi(\theta_2, k_2, q_2) - \pi] - [\phi(\theta_1, k_1, q_1) - \pi] = 0.$$

Thus, the inequality (B.1) becomes

$$\theta_2 - \theta_1 > -\frac{\phi_k}{\phi_\theta}[k_2 - k_1] - \frac{\phi_q}{\phi_\theta}[q_2 - q_1] = \theta_k[k_2 - k_1] + \theta_q[q_2 - q_1] \quad (\text{B.2})$$

where $\theta_k = -\phi_k/\phi_\theta$ and $\theta_q = -\phi_q/\phi_\theta$ by applying the implicit theorem on Equation (8). The

assumption 2 together with (B.2) yields

$$\begin{aligned} f(\theta(k_2, q_2, \pi)) - f(\theta(k_1, q_1, \pi)) &> f_\theta[\theta_2 - \theta_1] \\ &> f_\theta \theta_k [k_2 - k_1] + f_\theta \theta_q [q_2 - q_1] \end{aligned} \quad (\text{B.3})$$

By setting $k_1 = 0$ and $q_1 = 0$, the inequality (B.3) becomes

$$\begin{aligned} F(k_2, q_2, \pi) &> F_k k_2 + F_q q_2 + F(0, 0, \pi) \\ &> F_k k_2 + F_q q_2. \end{aligned}$$

Thus, $F - F_k k^* - F_q q^* > 0$. Applying the same argument to the composite function $H(k, q, \pi) = h(\theta(k, q, \pi), \pi)$ yields $H - H_k k^* - H_q q^* > 0$. ■

C Benchmark Functional Forms and Numerical Mapping

The theoretical model above is intentionally stated under general monotonicity and concavity assumptions. The minimum-distance benchmark in Section 6, therefore, requires benchmark-specific parametric choices. This appendix records those choices explicitly.

For the transaction-cost block, the benchmark uses

$$\phi(\theta, k, q) = A_\phi \frac{\theta}{1 - \theta} \exp[-a_k k - a_q q(2 - q)],$$

where $A_\phi > 0$, $a_k > 0$, and $a_q > 0$. Substituting this expression into Equation (8) gives the benchmark transaction cut-off

$$\bar{\theta}(k, q, \pi) = \frac{(\pi/A_\phi) \exp[a_k k + a_q q(2 - q)]}{1 + (\pi/A_\phi) \exp[a_k k + a_q q(2 - q)]}.$$

For $q \in [0, 1]$, this specification implies $\phi_\theta > 0$, $\phi_k < 0$, and $\phi_q < 0$, so higher transaction experience and a larger dollar pricing index both reduce the effective transaction cost of using dollars.

For the accumulation of dollarisation capital, the benchmark uses

$$f(\bar{\theta}) = a_f \bar{\theta}^{\alpha_f},$$

where $a_f > 0$ and $\alpha_f \in (0, 1)$. Hence f is strictly increasing and strictly concave on $[0, 1]$, in line with Assumption 2.

For the firm-side quotation block, the benchmark assumes that the idiosyncratic maintenance

cost is exponentially distributed,

$$G(\xi) = 1 - e^{-\gamma\xi}, \quad \xi \geq 0, \quad \gamma > 0.$$

Combining this distribution with the threshold rule in Appendix B yields

$$h(\bar{\theta}, \pi) = \kappa_q \left[1 - \exp \left(-\gamma \max\{\mu\pi - \omega(1 - \bar{\theta}), 0\} \right) \right].$$

This function is increasing in both $\bar{\theta}$ and π ; on the region where the threshold is positive, it inherits concavity from the exponential distribution and is strictly concave in $\bar{\theta}$ for fixed π .

Finally, because Figure 10 reports deposit dollarisation rather than a pure transaction-share object, the benchmark maps the model into the observed series through the composite

$$D_t = \nu_D q_t + (1 - \nu_D) \bar{\theta}_t, \quad \nu_D = 0.48.$$

This measurement bridge is used only for the minimum-distance comparison in Section 6; it is not part of the theoretical model itself. It should be interpreted as a reduced-form proxy relationship, consistent with the cross-country evidence in Drenik and Perez (2021) and the Peruvian pricing-law evidence in Castellares and Toma (2019). The common numerical values used in the benchmark are

$$A_\phi = 0.1391, \quad a_k = 1.0315, \quad a_q = 2.2442, \quad a_f = 0.0996, \quad \alpha_f = 0.6762, \quad \delta_k = 0.1154, \quad \mu = 2.0267,$$

$$\omega = 0.7920, \quad \gamma = 4.3832, \quad \kappa_q = 0.2937, \quad \delta_q = 0.0005, \quad \eta^W = 0.3138, \quad \eta^S = 0.4781, \quad k_0 = 24.91.$$

The policy wedge itself follows the dated schedule implied by Table 2: Cambodia keeps $\eta_t = 0$, Laos P.D.R. moves from η^W to η^S in 2008, Vietnam moves from η^W to η^S in 2011, and Peru moves from 0 to η^S in 2004. Countries therefore differ through the depreciation path fed into the model, the policy schedule η_t , and the inherited stock q_0 of dollar quotations. The minimum-distance objective matches the anchored deposit-dollarisation paths in Figure 10 together with the two Cambodian quotation-share moments reported in the text (28% in 2010 and 43% in 2014). Numerically, the minimisation is implemented with a bounded reparameterisation, a coarse random search of 180 draws, and 6 local Nelder-Mead refinements from the best starting values, using random seed 1.

D Robustness of the Measurement Bridge

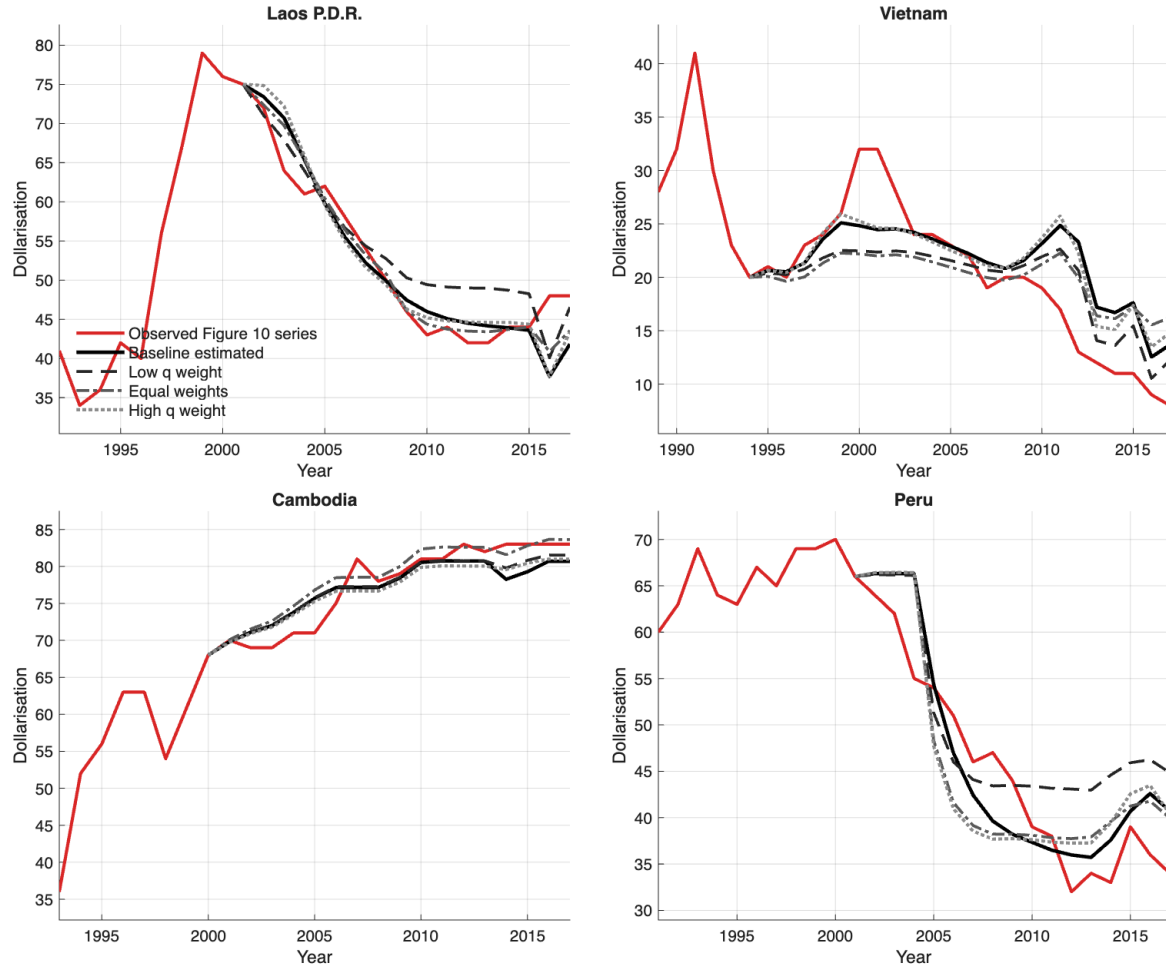
Table D1 evaluates whether the benchmark conclusions are sensitive to the particular weight assigned to q_t in the composite observable $D_t = \nu_D q_t + (1 - \nu_D)\bar{\theta}_t$. In each robustness exercise, the benchmark is re-estimated under the same dated policy schedules and the same Cambodian quotation-share moments, while the weight ν_D is either estimated freely or fixed at alternative values. The purpose is not to reopen the choice of cross-country outcome, which remains deposit dollarisation throughout, but to verify that the empirical support for the quotation channel is not driven by a single measurement bridge.

Table D1: Robustness to alternative measurement-bridge weights

Scenario	ν_D	Ext. RMSE	Uribe RMSE	Cambodia q RMSE	Wins
Baseline estimated weight	0.48	4.03	19.34	7.66	4/4
Low weight on q_t	0.35	4.68	19.34	7.33	4/4
Equal weights	0.50	4.27	17.06	8.92	4/4
High weight on q_t	0.65	4.42	19.37	7.38	4/4

Note: Each row re-estimates the benchmark under the same depreciation paths, the same dated policy schedules from Table 2, and the same Cambodian quotation-share moments, while varying only the weight on q_t in the observable $D_t = \nu_D q_t + (1 - \nu_D)\bar{\theta}_t$. “Ext. RMSE” and “Uribe RMSE” are the root mean squared deviations between the anchored model paths and the observed Figure 10 deposit-dollarisation series across the four benchmark countries. “Wins” reports the number of countries in which the extended model fits better than Uribe on that criterion.

Figure D1: Robustness of benchmark paths to alternative measurement bridges



Note: The red line is the original Figure 10 deposit-dollarisation series. The black and grey lines are the re-estimated extended-model paths under the baseline estimated bridge and the three alternative fixed values of ν_D . The depreciation paths are omitted because they are unchanged across these robustness exercises.

The robustness exercise supports the main interpretation of Section 6. Across all four variants, the extended model continues to outperform the Uribe benchmark in each country when fit is evaluated on the anchored deposit-dollarisation paths. The quantitative fit varies somewhat across bridges, as expected, but the overall conclusion remains stable: Cambodia, Laos P.D.R., and Peru remain the cases in which the quotation channel is most empirically valuable, while Vietnam remains the hardest case.

Figure D1 makes the same point visually. In Cambodia, all four bridge specifications keep the extended-model path close to the observed high-dollarisation trajectory, whereas the Uribe benchmark in Table D1 remains far below the data. In Laos P.D.R. and Peru, the alterna-

tive bridge specifications shift the precise speed of de-dollarisation but leave the basic ranking unchanged: the extended model produces a much steeper decline than Uribe and remains substantially closer to the observed path. Vietnam is again the least well-fitted country. There, the alternative bridge specifications shift the simulated path somewhat, but none overturn the broader conclusion that Vietnam is the hardest quantitative case for the model.

More specifically, the estimated benchmark weight $\nu_D = 0.48$ is close to the equal-weights case, and the overall fit of the extended model changes only modestly when ν_D is fixed at 0.35, 0.50, or 0.65. In the low-weight case, the extended model tracks the end-of-sample observations more closely in Laos P.D.R. and Vietnam, but it fits Peru less well. In the high-weight case, the Cambodia and Peru paths remain close to the baseline, while Vietnam again remains the weakest country. The equal-weights case is the most favourable variant for the Uribe benchmark, particularly in Cambodia, yet even there the extended model remains closer to the observed country paths overall. Thus, the paper’s main message does not depend on a knife-edge choice of the measurement bridge.

The robustness check should nevertheless be read with the same caution as the baseline benchmark. Varying ν_D changes only the empirical mapping from the model states into deposit dollarisation; it does not convert the benchmark into a full structural estimation of deposits, payments, and quotation behaviour jointly. The value of the exercise is narrower but still important: it shows that the empirical support for the quotation channel is stable under reasonable alternative ways of linking the model to the common Figure 10 outcome.